Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Ripple Effect**
b. Team leader name: **Victor Chang**
c. Problem Statement:

Supplier changes, such as a discontinued part or a jump in copper and freight costs, take days and six teams to trace to products, margins and customers. Manufacturers need that answer, and a plan, before the first meeting.

Brief about the idea

One supplier notice in, one approved response plan out

RippleOps is an AI change-impact agent for manufacturers. When a supplier discontinues a part, or raw-material and freight costs move, it:

- **Traces** every level and path of the bill of materials, with BigQuery Graph
- **Prices** cost, margin, revenue at risk and stock coverage, deterministically
- **Responds** with nine Google ADK agents that source alternatives and re-plan production
- **Hands off** owned actions to every team; a person approves the plan

DEMO · NOVACHIP ENDS MCU-X100

12 affected SKUs in 3 product families	**$4.2M** revenue exposed over the next 6 months
58 days of MCU-X100 inventory left	**4** products with no approved alternative

Fictional seeded dataset · figures asserted by tests

Opportunities

From days of spreadsheets to one calculated, shared answer

How is it different?

- ERP and PLM store the BOM; they do not trace a change to margin and customers
- Chatbots guess numbers; RippleOps calculates them and checks every AI figure
- One flow across sourcing, planning and finance, not one team's view

How does it solve the problem?

- Gemini reads the notice; BigQuery Graph finds the true quantity per product
- Simulates swaps and cost moves, scores suppliers, re-plans 13 weeks
- Writes owned, prioritized actions for every team

USP

- AI reasons, engines calculate: figures a CFO can sign off on
- Nine Google ADK agents, streamed live, with a human approval step
- Live end to end on Google Cloud today

WHY NOW

52%
of electronic parts discontinued in 2025 had no end-of-life notice
Z2Data, 2026

45%
of one year's profits lost to supply disruptions per decade
McKinsey Global Institute, 2020

26,225
supply chain disruption alerts in 2025, up from 22,522
Resilinc EventWatch, 2025

$210B
auto revenue lost to the 2021 chip shortage
AlixPartners, 2021

List of features offered by the solution

Eight features, one flow from notice to decision

Notice reader

Gemini turns a PDF, TXT or DOCX supplier notice into a structured event

Multi-level BOM trace

BigQuery Graph walks every path to the true quantity per product

Impact dashboard

Affected SKUs, revenue exposure, stock coverage and a Why? panel

Scenario simulator

Swap parts, move copper and freight, set a margin target

Alternative sourcing

Suppliers scored 0–100 on six factors behind a sanctions gate

Forecast and re-plan

AI.FORECAST (TimesFM) demand and a 13-week OR-Tools re-plan

Nine-agent response

Google ADK agents run in sequence and in parallel, streamed live

Actions and approval

Owned actions per team, each citing its facts; a person approves

Process flow diagram or Use-case diagram

Seven steps from supplier notice to approved plan

1 CHANGE	2 TRACE	3 CALCULATE	4 SOURCE	5 RE-PLAN	6 DECIDE	7 ACT
Gemini reads the supplier notice	BigQuery Graph walks every BOM level	Cost, margin, revenue and stock coverage	Agents vet and score alternate suppliers	TimesFM forecast, OR-Tools re-plan	A planner approves the proposed plan	Owned actions for every team

IN THE DEMO · MCU-X100 END-OF-LIFE

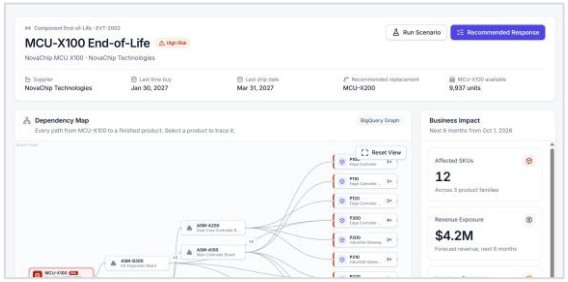
NovaChip ends MCU-X100; last-time buy 30 Jan 2027	12 affected SKUs in 3 product families	$4.2M revenue exposed; 58 days of stock	5 options scored; 1 blocked by sanctions	Bridge buy of 500 units; fill rate 64% → 100%	Plan PLAN-EVT-2001-02 approved	Procurement, engineering, finance and sales

Gemini agents (Google ADK) · Deterministic engines (BigQuery, Python, OR-Tools) · A person

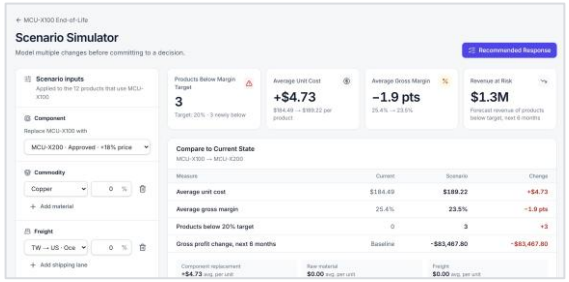
One flow: under two minutes of agent work (18.5 s on the latest live run) turns a supplier notice into a plan a person can approve.

Wireframes/Mock diagrams of the proposed solution (optional)

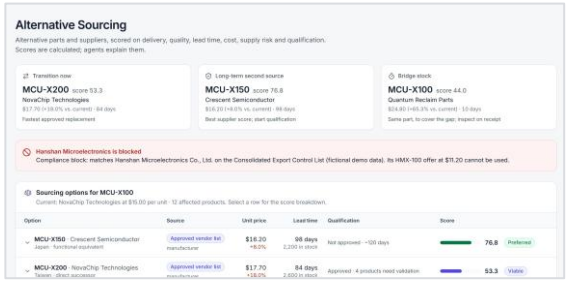
Six live screens, one case: the planner's path through RippleOps



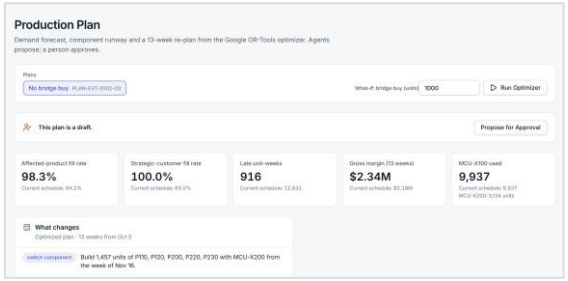
1 Impact Analysis the whole ripple



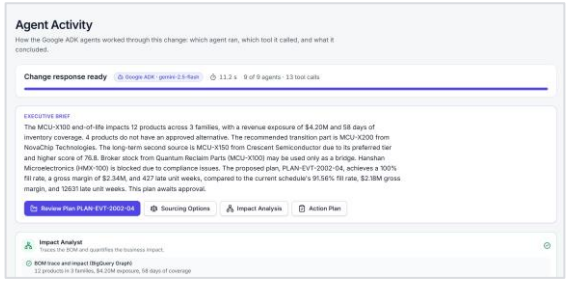
2 Scenario Simulator test a decision



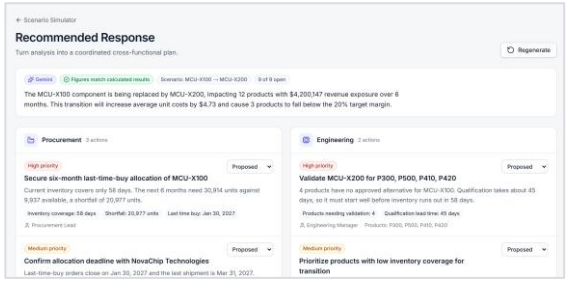
3 Alternative Sourcing suppliers scored



4 Production Plan 13-week re-plan



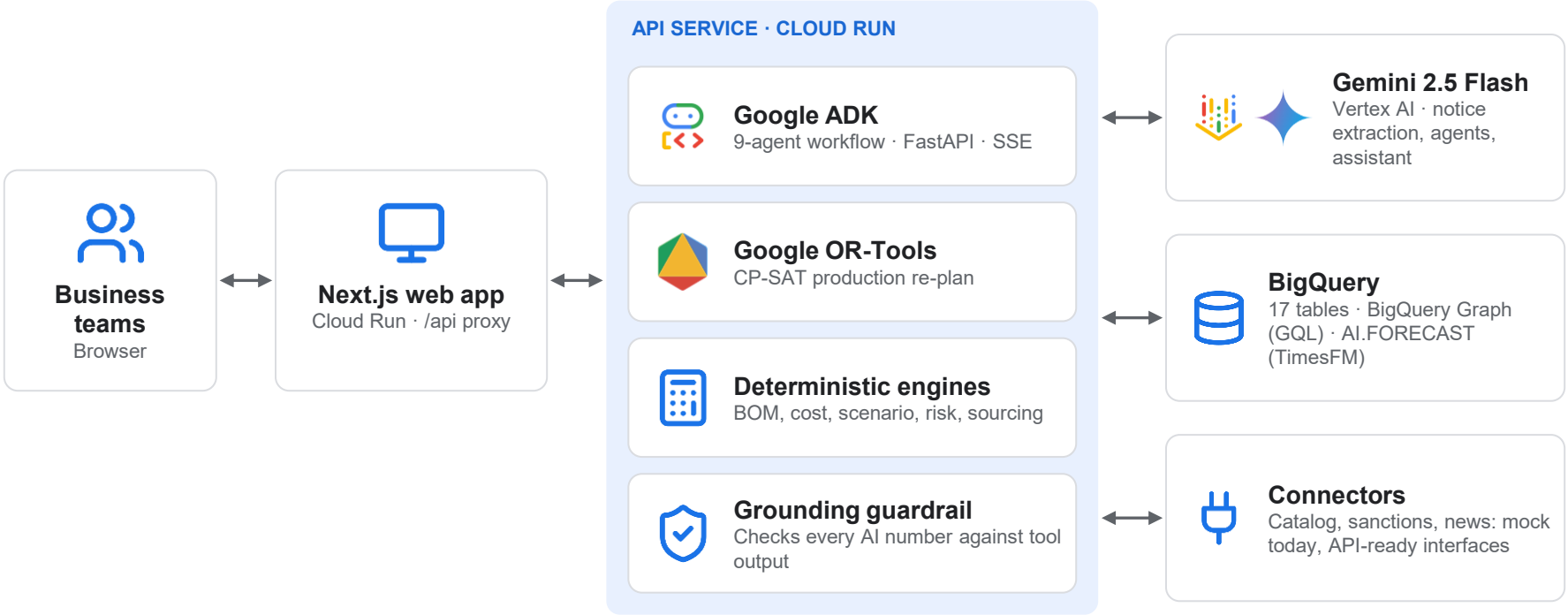
5 Agent Activity nine agents, live



6 Recommended Response next moves

Architecture diagram of the proposed solution

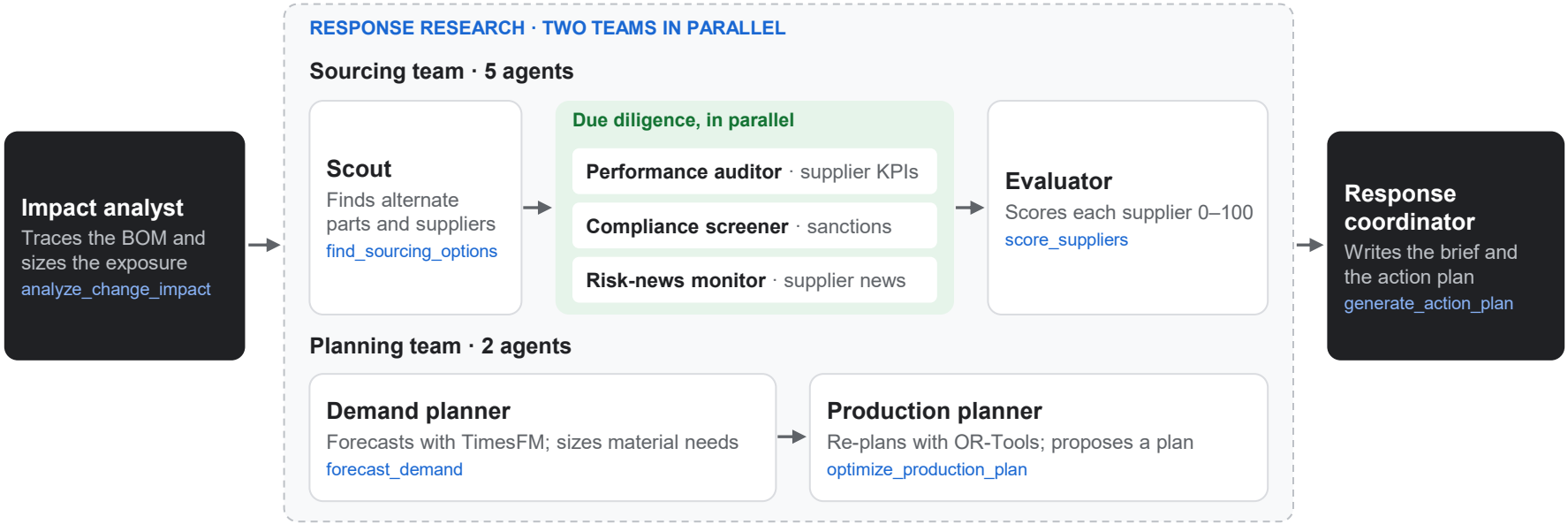
Built end to end on Google Cloud and Google AI



Product logos © Google, used to identify Google services. Connectors use fictional fixtures; DigiKey, Mouser, OpenSanctions or a news API plug into the same contract.

Gen AI implementation · Google ADK multi-agent workflow

Nine AI agents work the case, two teams in parallel



Agents share findings through ADK session state

Every agent and tool step streams live to the UI

The production plan is only proposed; a person approves it

Trustworthy by design · AI reasons, engines calculate

Numbers your CFO can sign off on

Gemini via Google ADK: reasons	BigQuery, Python and OR-Tools: calculate
Reads supplier notices into structured events	Multi-level BOM trace with BigQuery Graph
Chooses which tools to call, and in what order	Cost roll-up, gross margin and scenarios
Decides whether a bridge buy is worth testing	Production re-plan with OR-Tools CP-SAT
Explains forecasts, scorecards and plans	Demand forecast with BigQuery AI.FORECAST
Writes findings, the brief and action wording	Supplier scores, compliance gate, risk levels



Grounding guardrail

An ADK after-model callback checks every number an agent writes against tool output. Anything ungrounded falls back to the calculated finding. Without Gemini, the same tools still run in order.

Technologies to be used in the solution

Google Cloud and Google AI at the core, open source around it

Google AI

Gemini 2.5 Flash on Vertex AI

Google ADK 2.9 multi-agent workflow

AI.FORECAST TimesFM in BigQuery

Data

BigQuery 17 business tables

BigQuery Graph GQL BOM trace

Recursive SQL automatic fallback

Optimization and engines

OR-Tools 9.15 CP-SAT re-plan

Python 3.11 cost, scenario, risk

Grounding check AI numbers vs tools

Backend

FastAPI REST and server-sent events

Pydantic 2 typed tools and schemas

pypdf notice parsing

Frontend

Next.js 16, React 19 TypeScript

React Flow dependency map

Recharts, Tailwind 4 charts and UI

Cloud and quality

Cloud Run web and API services

Cloud Build source deploys

pytest 92 automated tests

Estimated implementation cost (optional)

A one-site pilot runs for roughly $50–120 a month

Component	Sizing	Est. / month
Cloud Run · API	1 vCPU, 2 GiB, 1 warm instance	**$20–60**
Cloud Run · web	1 vCPU, 512 MiB, scales to zero	**< $5**
BigQuery storage and queries	17 tables, MB-scale data	**< $5**
BigQuery Graph slots	Enterprise, baseline 0, autoscale ≤ 50	**$10–30**
Gemini 2.5 Flash	~200 change events, 9 agents each	**$10–20**
Total, one-site pilot		**≈ $50–120**

Estimates from Google Cloud list prices (us-central1) and the current deployment's sizing, not a quote. Validate with the Google Cloud Pricing Calculator.

≈ $0.10–0.25

per change event analyzed
Gemini tokens for nine agents plus BigQuery Graph slot time

$0

idle BigQuery slot cost
Reservation baseline is 0 slots; it bills only while queries run

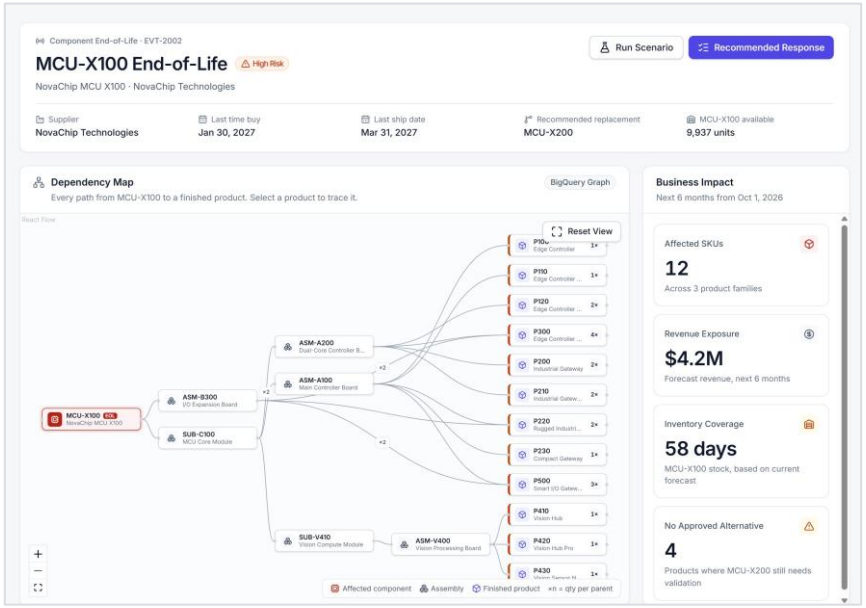
$0

software licences
ADK, OR-Tools, FastAPI and Next.js are open source

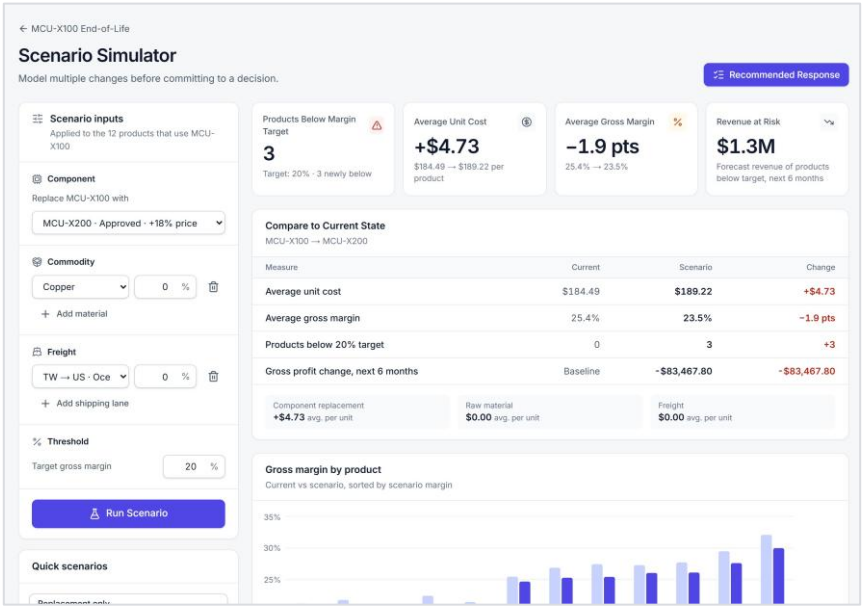
Snapshots of the prototype

Live on Google Cloud · fictional demo data

See the whole ripple, then test a decision



Impact Analysis
BigQuery Graph traces every path from MCU-X100 to 12 products

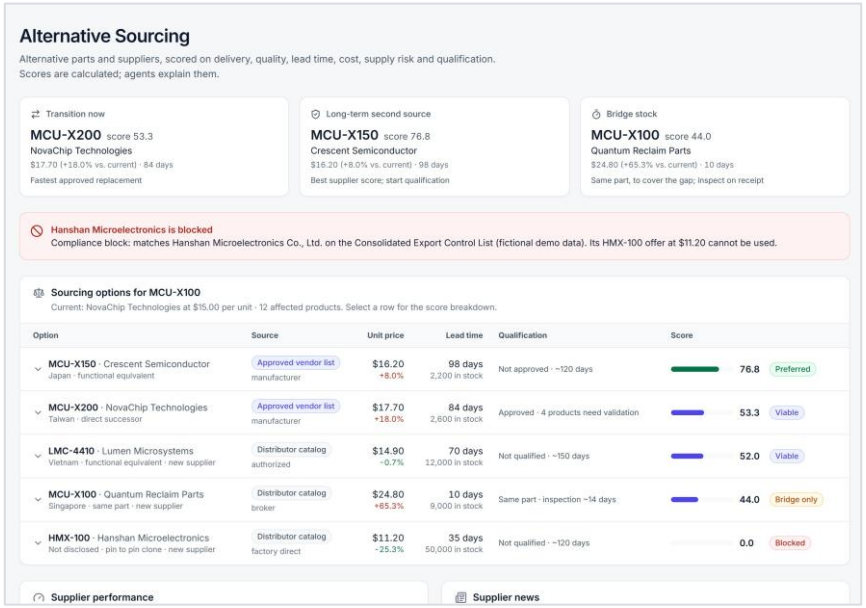


Scenario Simulator
MCU-X200: +$4.73 unit cost, −1.9 pts margin, 3 below target

Snapshots of the prototype

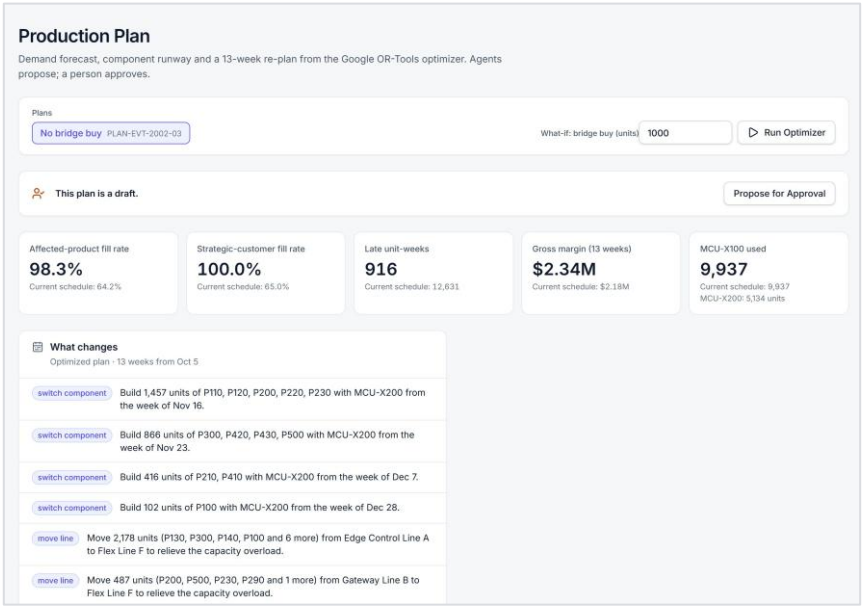
Live on Google Cloud · fictional demo data

Find the next supplier, then re-plan the line



Alternative Sourcing

Five options scored 0–100; one blocked by a sanctions match



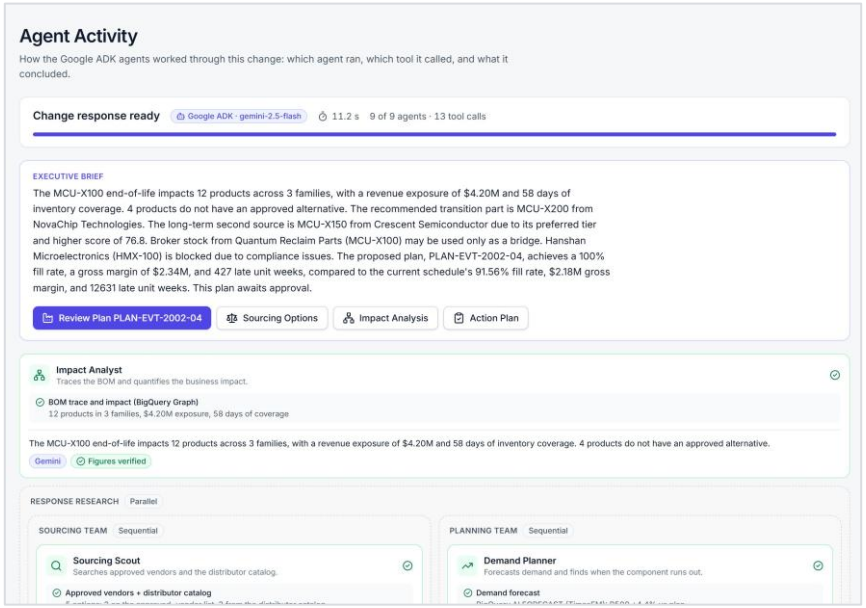
Production Plan

OR-Tools re-plans 13 weeks; the plan waits for a person

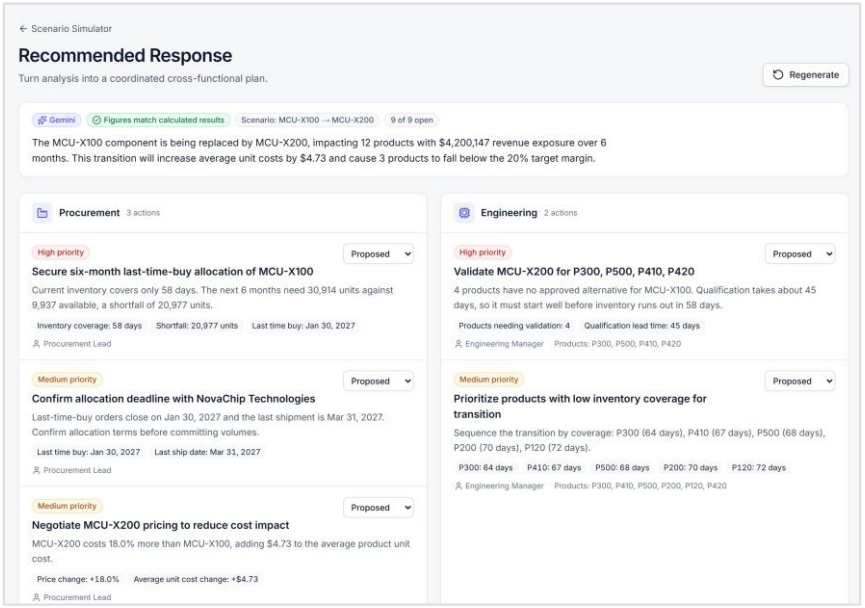
Snapshots of the prototype

Live on Google Cloud · fictional demo data

Nine agents brief every team on its next move



Agent Activity
9 of 9 ADK agents finished; executive brief with verified figures



Recommended Response
Prioritized actions per team, each citing its facts

Prototype Performance report/Benchmarking

Fast enough for a live meeting, verified by 92 tests

18.5 s

Full nine-agent run
Latest live run on Gemini 2.5 Flash; 13–94 s observed

13

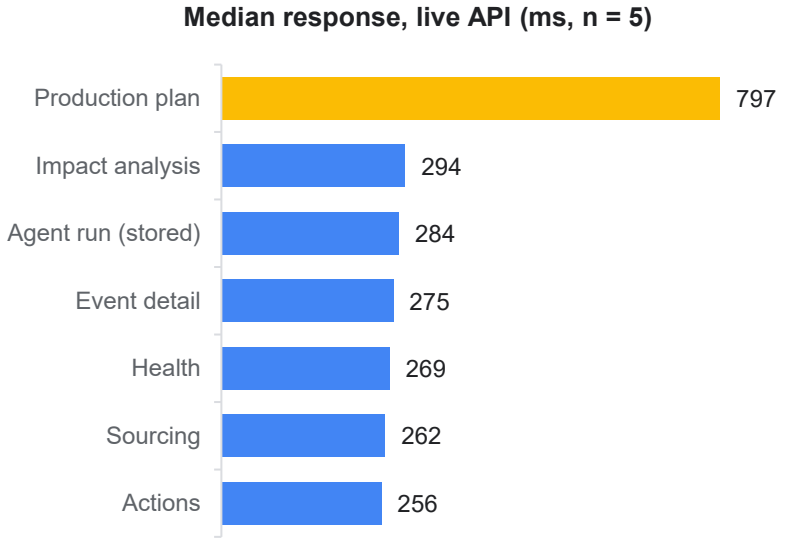
Tool calls per run
56 agent and tool steps streamed live to the UI

< 0.3 s

Median API response
Impact, sourcing and actions on Cloud Run

92 / 92

Automated tests pass
Unit, integration and API tests, in 15 s



Accuracy: demo figures (12 SKUs, $4.2M, 58 days, +$4.73, −1.9 pts) are asserted in CI; the BigQuery Graph trace matches recursive SQL and the seed.

Additional Details/Future Development (if any)

From prototype to an always-on supply chain co-pilot



NEXT

Connect

- ERP and PLM connectors for BOMs, stock and orders
- Real catalog, sanctions and news APIs on today's interfaces
- Single sign-on and role-based access

THEN

Act

- Write approved actions back to ERP and ticketing
- Approval workflow and audit trail for every decision
- Notify owners by e-mail or chat

LATER

Watch

- Continuous monitoring of supplier feeds
- Tariff, commodity, freight and supplier-failure triggers
- Multi-plant, multi-region BOMs at BigQuery scale

Same agents and engines everywhere: electronics, automotive, industrial equipment and medical devices; only the data and connectors change.

Provide links to your:

Try RippleOps live, read the code, watch the demo

GitHub Public Repository
github.com/victorchangwork052/rippleops

Demo Video Link (3 Minutes)
https://drive.google.com/file/d/1c1dxjsODO5XLfD-MqT9WZJLhYdOSqMGH/view?usp=sharing

Final Product Link
rippleops-web-273776274235.us-central1.run.app

Try the live demo

1. Open the product link
2. Select Load Demo Scenario
3. Select Analyze Impact to start the nine-agent run

All demo data is fictional

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

